

AVL Unified Tools Portal

Product Requirements Document

Web Application Framework & Organigram Collaboration Feature

Field	Value
Document Version	0.1 Draft
Status	In Review
Date	June 2026
Product	AVL Unified Tools Portal
Author	Vijey R S
Organisation	AVL India

CONFIDENTIAL - INTERNAL USE ONLY

1. Overview

1.1 Purpose

The AVL Unified Tools Portal is an enterprise-grade web application that consolidates three previously standalone operational tools (LMM Planner, Organigram Creator, and Weekly Dashboard) into a single, secure, authenticated platform. This PRD covers the architectural framework for the unified portal and the introduction of a new collaborative Access Control List (ACL) Sharing feature for the Organigram Creator.

1.2 Background

Historically, the LMM Planner, Organigram Creator, and Weekly Dashboard existed as isolated, heavy-DOM-manipulation HTML/JS files. Data persistence relied on local JSON file imports and exports. This approach hindered cross-team collaboration, lacked data security, and resulted in fragmented workflows.

This product addresses these issues by:

- Transitioning the standalone tools into an Angular Single Page Application (SPA) wrapper.
- Introducing a FastAPI backend with a SQLite database to replace manual file handling with secure cloud persistence.
- Enabling seamless sharing of Organigrams between authenticated colleagues.

1.3 Scope

In Scope

- User authentication (Login, Registration, JWT sessions).
- A centralized App Hub ("My Files" and "Shared with Me" dashboards).
- Angular wrapper integration for the existing Vanilla JS tools.
- Organigram collaboration feature (View/Edit ACL sharing).
- Database persistence for LMM Planner, Organigram Creator, and Weekly Dashboard payloads.
- Export capabilities (PNG, PDF/Print, Excel) per tool specifications.

Out of Scope

- A complete rewrite of the tools' internal vanilla DOM manipulation into strictly idiomatic Angular (Phase 2 future initiative).
- Real-time concurrent collaborative editing (e.g., Google Docs style cursor tracking).
- Enterprise SSO integration (Active Directory/SAML) for the MVP.

2. Goals

- **Centralize Workflow:** Provide a single launchpad for AVL teams to access project planning, organizational mapping, and weekly tracking.
- **Ensure Data Integrity & Security:** Replace manual JSON file transfers with a secure, backed-up SQLite database protected by user authentication.
- **Enable Collaboration:** Allow users to securely share Organigrams with colleagues directly within the app without sending attachments.

- **Minimize Migration Friction:** Wrap existing, proven vanilla JS logic within Angular to accelerate delivery while laying the groundwork for a future UI revamp.

3. User Personas

Persona	Role / Needs
Project Manager	Creates LMM Planners and Weekly Dashboards. Uploads and manages baseline schedules. Needs quick access to all owned files.
Team Leader	Creates Organigrams. Uses the new "Share" feature to distribute organizational structures to relevant team members.
Team Member	Logs in primarily to view Organigrams or Dashboards shared with them via the "Shared with Me" hub.
System Admin	Manages the database, oversees user registrations, and monitors system health.

4. System Components

Layer	Technology	Role
Frontend	Angular, HTML, CSS	Application Hub, routing, auth guards, and component wrappers for legacy Vanilla

Layer	Technology	Role
		JS tools.
Backend	FastAPI (Python), Pydantic	REST API, user authentication, business logic, payload routing, and ACL management.
Database	SQLite, SQLAlchemy	Relational data storage for users, file metadata, sharing permissions, and JSON payloads.
Security	JWT (JSON Web Tokens)	Stateless session management and route protection.

5. Key Features

5.1 Authentication & Central Launchpad (App Hub)

Users log in to access a central dashboard. The Hub displays options to launch any of the three tools and presents two distinct file lists: "My Files" (owned by the user) and "Shared with Me" (owned by others, shared via ACL).

5.2 LMM Planner

Maintains its full feature set (Gantt charts, Critical Path Method, baseline tracking, resource management, and milestone tracking). The "Load/Save JSON" functionality is seamlessly redirected to backend API calls.

5.3 Organigram Creator & Sharing Workflow

Users can build organizational charts with flexible coordinators and team positions. A new "Share" button allows the owner to input a colleague's email address and assign a permission level (Viewer or Editor). The chart immediately appears in the recipient's Launchpad. Standard exports (PNG, Print/PDF) remain fully functional.

5.4 Weekly Dashboard

The operational view for macro project indicators. Supports multi-week history, auto-derived global trajectories based on weighted items, and Excel export/import, with data natively syncing to the unified database.

6. Functional Requirements

6.1 Authentication & Hub

ID	Requirement	Priority
FR-01	Users shall be able to register and log in using an email and password.	Must Have
FR-02	The system shall issue a JWT upon successful login for session management.	Must Have
FR-03	The App Hub shall display a "My Files" list containing all projects owned by the user.	Must Have
FR-04	The App Hub shall display a "Shared with Me" list containing projects shared with the user.	Must Have

6.2 Organigram Collaboration (New Feature)

ID	Requirement	Priority
FR-05	The Organigram tool shall include a "Share" button triggering a sharing modal.	Must Have
FR-06	The sharing modal shall allow the owner to input a registered user's email.	Must Have
FR-07	The owner shall be able to assign "View" or "Edit" permissions to the invitee.	Must Have
FR-08	Shared organigrams shall immediately populate in the invitee's "Shared with Me" hub.	Must Have

6.3 Tool Wrappers & Persistence

ID	Requirement	Priority
FR-09	Existing tools shall render within Angular components using <code>ngAfterViewInit</code> .	Must Have
FR-10	Tools shall export their state as JSON payloads to the FastAPI backend on "Save".	Must Have
FR-11	Tools shall import their state by fetching JSON payloads from the FastAPI backend on load.	Must Have
FR-12	Organigram shall retain PNG and Print/PDF export capabilities locally.	Must Have

7. User Flows

7.1 Platform Authentication Flow

1. User navigates to the web application URL.
2. User enters credentials at the login screen.
3. FastAPI validates credentials and returns a secure JWT.
4. Angular stores the JWT and redirects the user to the App Hub.
5. User views their saved files and selects a tool to launch or a file to open.

7.2 Organigram Sharing Flow

1. User opens an existing Organigram from the App Hub.
2. User clicks the "Share" button located in the top toolbar.
3. An Angular modal overlays the tool interface.
4. User enters a colleague's email and selects "Viewer" or "Editor" access.
5. User clicks "Confirm". FastAPI updates the Organigram_Shares table.
6. The colleague logs in, checks their "Shared with Me" tab, and opens the Organigram with the granted permissions.

8. Data Model

8.1 Users

Column	Type	Description
id	INTEGER PK	Unique identifier for the user
email	VARCHAR(255)	User's email address (Unique)
hashed_password	VARCHAR(255)	Bcrypt hashed password

8.2 Organigrams

Column	Type	Description
id	INTEGER PK	Unique identifier for the chart
owner_id	INTEGER FK	References Users.id
name	VARCHAR(200)	Name of the Organigram
json_payload	TEXT	Complete JSON state of the chart
updated_at	DATETIME	Last modification timestamp

8.3 Organigram_Shares (ACL)

Column	Type	Description
share_id	INTEGER PK	Unique identifier for the share record
org_id	INTEGER FK	References Organigrams.id
shared_with_user_id	INTEGER FK	References Users.id
permission_level	VARCHAR(50)	'viewer' or 'editor'

8.4 LMM_Projects & Dashboards

Column	Type	Description
id	INTEGER PK	Unique identifier
owner_id	INTEGER FK	References Users.id
name	VARCHAR(200)	Project or Week identifier
json_payload	TEXT	Complete JSON state of the respective tool

9. API Endpoints

9.1 Authentication & Hub

Method	Endpoint	Description
POST	/api/users/register	Create a new user account
POST	/api/users/login	Authenticate and return JWT
GET	/api/my-files	Return list of owned and shared files across all tools

9.2 Organigram & Sharing

Method	Endpoint	Description
GET	/api/org/{id}	Return Organigram payload (requires ownership or ACL permission)
POST	/api/org	Create a new Organigram record
PUT	/api/org/{id}	Update existing Organigram payload
POST	/api/org/{id}/share	Grant access to a specific user email

9.3 Planner & Dashboard

Method	Endpoint	Description
GET	/api/lmm/{id}	Return LMM Planner payload
PUT	/api/lmm/{id}	Update LMM Planner payload
GET	/api/dash/{id}	Return Weekly Dashboard payload
PUT	/api/dash/{id}	Update Weekly Dashboard payload

10. Non-Functional Requirements

- **Security:** All API routes (except login/register) must be protected by JWT validation. Cross-Origin Resource Sharing (CORS) must be strictly configured.
- **Performance:** The Angular wrapper must load the vanilla DOM manipulation scripts without degrading rendering speed.
- **Database:** SQLite will be used for the MVP. The SQLAlchemy ORM must be configured to easily migrate to PostgreSQL if enterprise scaling is required in the future.
- **Compatibility:** Export features (specifically html2canvas for PNGs) must remain fully functional inside the Angular component lifecycle.

11. Open Items

#	Open Item	Status
1	Determine if the collaborative "Share" feature should be immediately extended to the LMM Planner and Weekly Dashboard, or monitored in Organigram first.	Open
2	Finalize the timeline for Phase 2: rewriting the vanilla JS interfaces into native Angular RxJS structures.	Open
3	Confirm email service integration requirements for notifying users when an Organigram is shared with them.	Open